Heme Chemistry

Hemoglobin

- Hemoglobin is red blood pigment, found in erythrocytes
- It is a chromoprotein, containing heme as the prosthetic group & globin as the protein part-apoprotein
- Heme containing proteins are characteristic of aerobic organisms
- Normal levels
- Adult male: 14 to 16 gm%
- Female: 13 to 15 gm%

- Hemoglobin is a tetrameric protein & molecular weight 64,450
- Approximately 6.25 gm of Hb are produced & destroyed in body each day
- The basic protein "globin" varies from species to species in its amino acid composition & sequence, and is responsible for species-specificity
- Polypeptide chains of globin of adult Hb, contain high content of 'histidine' & 'lysine' & small amount of isoleucine

Functions of Hemoglobin

- Delivery of O_2 from lungs to the tissues
- Transport of CO_2 & protons from tissues to lungs for excretion
- Heme is present in Myoglobin, Cytochromes, Peroxidase, Catalase, Tryptophan pyrrolase & Nitric oxide synthase
- In cytochromes, oxidation & reduction of iron is essential for their biological function in ETC

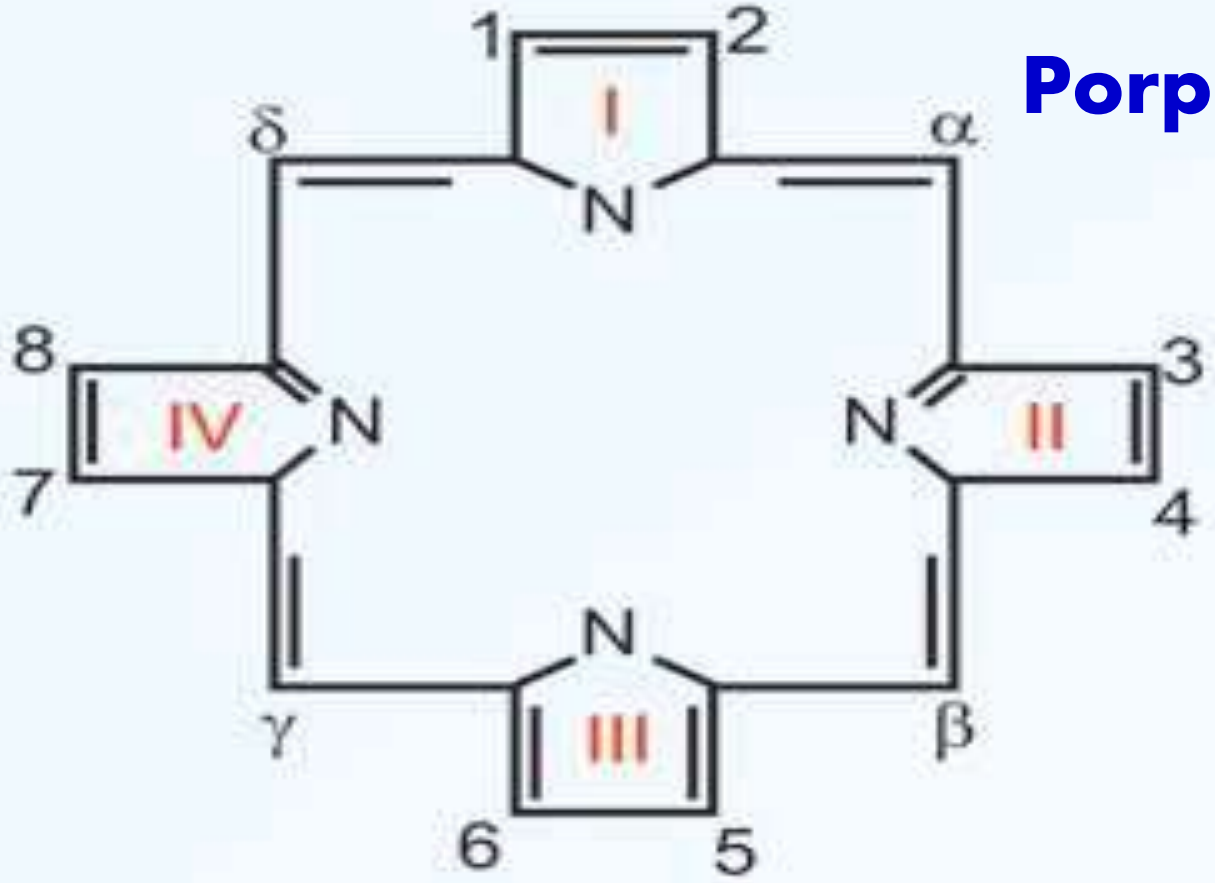
Structure of Globin

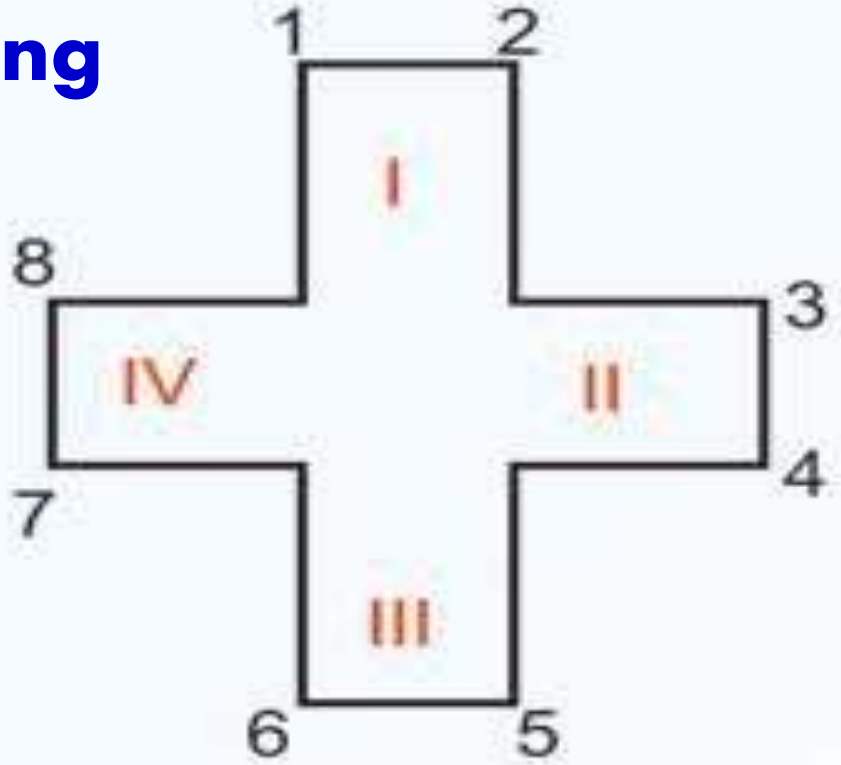
- Globin consists of 4 polypeptide chains
- Adult Hb is made up of 2α-chains & 2β-chains (α2 β2)
- Each α -chain contains 141 AAs & β-chain contains 146 AAs.
- HbA_1, has a total of 574 amino acids
- The four subunits of hemoglobin are held together by non-covalent interactions - hydrophobic, ionic & hydrogen bonds.
- Each subunit contains a heme group

Structure of Heme

- Heme is a Fe-porphyrin compound
- Porphyrins are cyclic compounds formed by fusion of 4 pyrrole rings linked by methenyl (=CH–) bridges
- Since an atom of iron is present, heme is a ferroprotoporphyrin.
- The pyrrole rings are named as I, II, III, IV and the bridges as alpha, beta, gamma and delta

Porphyrin ring

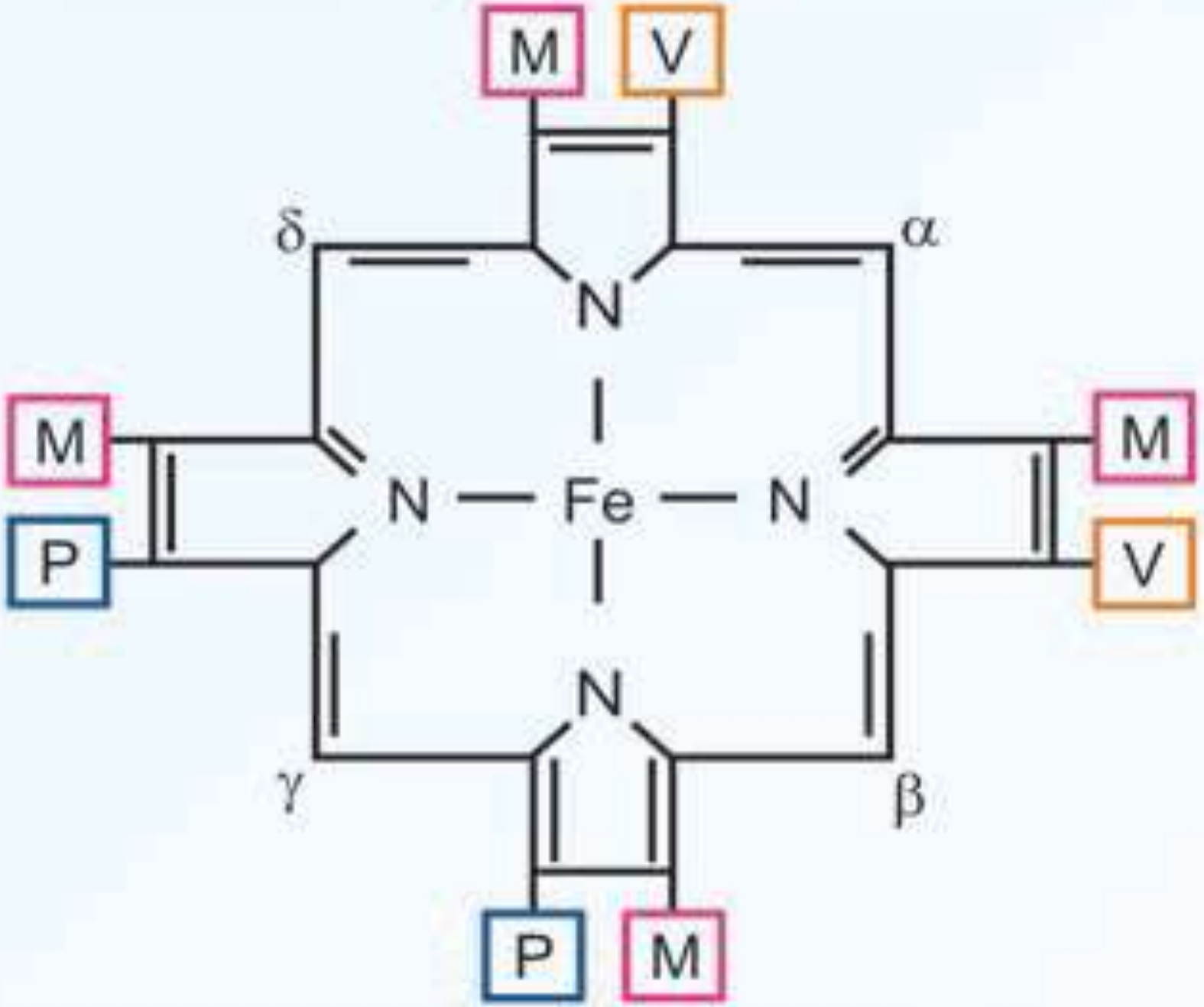




The pyrrole rings are numbered I to IV; the bridges named as alpha to delta and the possible sites of substitutions are denoted from 1 to 8. (For brevity, the bridges and double bonds are sometimes omitted, as shown on the right).

- Heme contains a porphyrin molecule, protoporphyrin IX, with iron at its center
- Protoporphyrin IX consists of four pyrrole rings to which four methyl, two propionyl & two vinyl groups are attached

Structure of heme



M = Methyl

V = Vinyl

P = Propionyl

Formation of Heme Pockets

- **Each polypeptide chain contains a 'heme pocket'**
- **Hb molecule & its sub-units contains hydrophobic amino acids internally & hydrophilic amino acids on their surfaces**
- **The *heme pockets of α-subunits are of size, adequate for entry of O_2 molecule, but the entry of O_2 into the heme-pockets of β-subunits is blocked by valine.***

Differences between α & β-chains of adult normal Hb

	α-Subunit	β-Subunit
Molecular weight	15126	15866
Total amino acids	141	146
C-terminal amino acid	Arginine	Histidine
N-terminal amino acid	Val-Leu	Val-His-Leu
α-Helices	7	8
Heme-pocket	Adequate for entry of one molecule of O_2	Entry of O_2 in heme-pocket is blocked by valine

Other forms of Haemoglobin

- **Hb-A_1:**
- **Normal adult Hb, commonly called Hb-A, consists of 2 α - & 2 β chains ($\alpha_2\beta_2$)**
- **It is approximately 90% of total haemoglobin**
- **Hb-F:**
- **It is a human foetal haemoglobin**
- **Consisting of $\alpha_2\gamma_2$**

Differentiation of Hb-A from Hb-F

Hb-A	Hb-F
Two α & two β chains	Two α & two γ chains
Denatured by alkali	Resistant to alkali denaturation
At pH 8.9 Hb-A moves ahead of Hb-F	Hb-F moves behind Hb-A
2,3-BPG content is high	2,3-BPG content is low
Affinity of O_2 is less	Affinity to O_2 is more
Delivery power of O_2 more (unloading)	Delivery power of O_2 is decreased
Concentration at birth-Hb-A=85%	15%
	Hb-F disappears by end of first year, persistence of Hb-F after one year is pathological

Hb-A_2:

- It is a minor component of normal adult Hb.
- It contains two α & two δ-chains α_2 δ_2
- It is approximately-2.5%
- Electrophoretically, it is a slowly migrating fraction

Hb-A_3:

- It amounts for 3 to 10% of total haemoglobin
- It is a fast moving fraction

Normal major types of haemoglobin

Type	Composition	% of total haemoglobin
HbA_1	$\alpha_2\beta_2$	90%
HbA_2	$\alpha_2\ \delta_2$	<5%
HbF	$\alpha_2\gamma_2$	<2%
HbA_{1c}	$\alpha_2\beta_2$-glucose	<5%

Hb-A_{1c} (Glycosylated Hb):

- It is formed by covalent binding of glucose to haemoglobin
- Its normal range is 3 to 6%
- Its levels are increased in diabetes mellitus
- **Chemistry:**
- The amino acid sequence of HbA_{1c} is exactly same as that of HbA_1
- The attachment of 1-amino 1-deoxy fructose to the $-NH_2$ terminal of valine of β-chain of HbA_1

- Addition of sugar moiety to valine occurs non-enzymatically, either by addition of glucose directly to the protein.
- Diagnostic importance of HbA_{1c}:
- The rate of synthesis of HbA_{1c} is directly related to the exposure of RBC to glucose
- The concentration of HbA_{1c} serves as an indication of blood glucose concentration over a period

- **HbA_{1c} reflects the mean blood glucose level over 3 months period prior to its measurement**
- **In diabetes, HbA_{1c} is elevated to as high as 15%**
- **Determination of HbA_{1c} is used for monitoring of diabetes**
- **If the HbA_{1c} concentration is <7%, the diabetic patient is considered to be in good control**

Myoglobin

- Myoglobin (Mb) is monomeric O_2 binding hemoprotein
- Found in heart and skeletal muscle.
- It has single polypeptide (153 A.As) chain with heme moiety.
- Myoglobin (mol. wt. 17,000) structurally resembles the individual subunits of hemoglobin molecule
- Myoglobin functions as a reservoir for oxygen.
- It serves as oxygen carrier that promotes the transport of oxygen to the rapidly respiring muscle cells

Binding of O_2 to haemoglobin

- One molecule of Hb can bind with four molecules of O_2.
- Myoglobin (with one heme) which can bind with only one molecule of oxygen.
- In other words, each heme moiety can bind with one O_2.

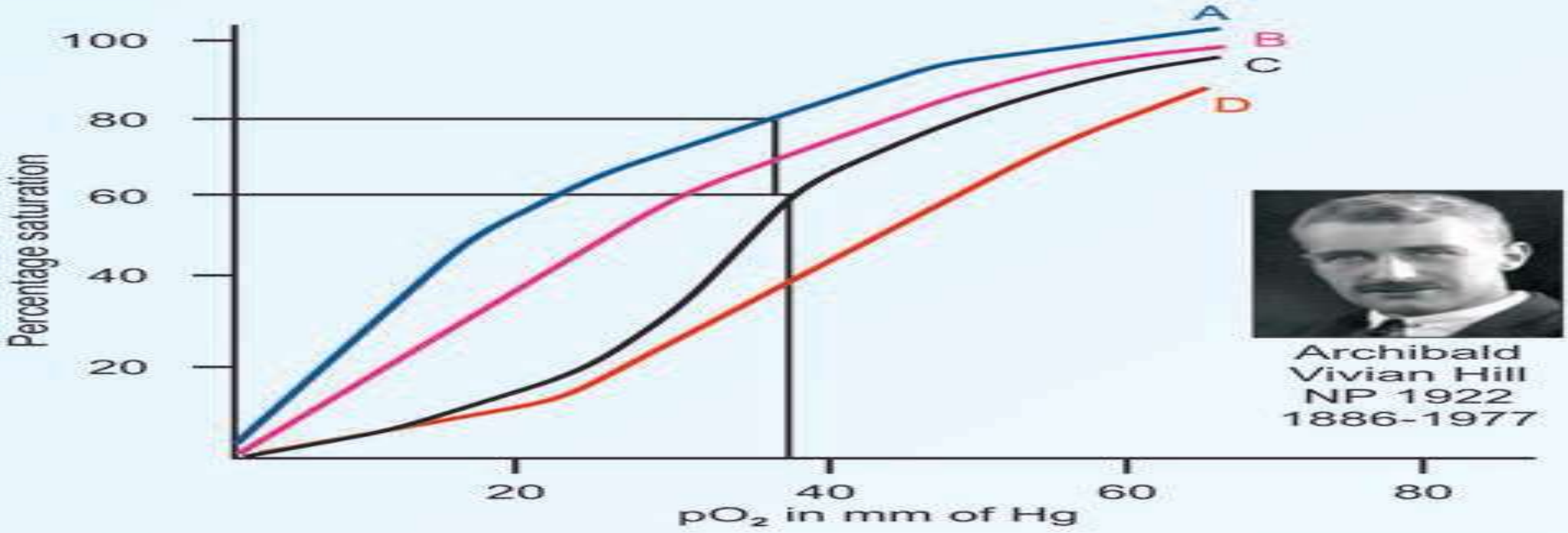
Transport of O_2 by haemoglobin

- It can transport large quantities of oxygen
- It can take up and release oxygen at appropriate partial pressures
- It is a powerful buffer.

Oxygen Dissociation Curve (ODC)

- The binding ability of hemoglobin with oxygen at physiological pO_2 (partial pressure of oxygen) is shown by the oxygen dissociation curve (ODC)
- At the oxygen tension in the pulmonary alveoli, the Hb is 97% saturated with oxygen.

Oxygen dissociation curve (ODC)



A—Theoretical curve as per mass action. **B**—Sigmoid curve, due to heme-heme interaction (Hill effect). **C**—Further shift to right due to carbon dioxide (Bohr effect) and BPG. This curve represents the pattern under normal conditions. **D**—Further shift to right when temperature is increased to 42°C.

Archibald Vivian Hill
NP 1922
1886-1977



Christian Bohr
(1855-1911)

Factors affecting oxygen dissociation curve

- **Heme-heme Interaction & Cooperativity:**
- **The oxygen dissociation curve (ODC) is sigmoid shape.**
- **The binding of O_2 to one heme residue increases the affinity of remaining heme residues for O_2.**
- **Thus the affinity of Hb for the last O_2 is about 100 times greater than the binding of the first O_2 to Hb.**
- **This is called positive cooperativity**

- **Release of O_2 from one heme facilitates the release of O_2 from others.**
- **The quaternary structure of oxy-Hb is described as R (relaxed) form; & deoxy- Hb is T (tight) form.**

$2\alpha + 2\beta$ (Deoxy-Hb – T-form) ⟶ $2\alpha,\beta$ (Oxy-Hb – R-form)

T and R forms of hemoglobin

- The four subunits ($\alpha_2\beta_2$) of hemoglobin are held together by weak forces.
- The relative position of these subunits is different in oxyhemoglobin compared to deoxyhemoglobin.
- T-form of Hb:
- The deoxy form of Hb exists in T or taut (tense) form.
- The H & ionic bonds limit the movement of monomers.
- The T-form of Hb has low oxygen affinity.

R-form of Hb

- The binding of O_2 destabilizes some of the hydrogen & ionic bonds particularly between $\alpha\beta$ dimers.
- This results in a relaxed form or R-form of Hb
- Therefore, the R-form has high oxygen affinity.

Transport of CO_2 by hemoglobin

- In aerobic metabolism, for every molecule of O_2 utilized, one molecule of CO_2 is liberated.
- Hemoglobin actively participates in the transport of CO_2 from the tissues to the lungs.
- About 15% of CO_2 carried in blood directly binds with Hb.
- The rest of the tissue CO_2 is transported as bicarbonate (HCO_3).

- CO_2 molecules are bound to the uncharged α-amino acids of hemoglobin to form carbamyl hemoglobin.
- The oxyHb can bind 0.15 moles CO_2/mole heme, whereas deoxyHb can bind 0.40 moles CO_2/mole heme.
- The binding of CO_2 stabilizes the T (taut) form of hemoglobin structure, resulting in decreased O_2 affinity for Hb.

Hemoglobin also helps in the transport of CO_2 as bicarbonate

- CO_2 enters the blood from tissues, the enzyme carhonic anhydrase present in erythrocytes catalyses the formation of carbonic acid (H_2CO_3).
- Bicarbonate (HCO_3^-) & proton (H^+) are released on dissociation of carbonic acid
- Hb acts as a buffer & immediately binds with protons

- Every 2 protons bound to Hb, 4 oxygen molecules are released to the tissues.
- In the lungs, binding of O_2 to Hb results in the release of protons.
- The bicarbonate & protons combine to form carbonic acid.
- Acted upon by carbonic anhydrase to release CO_2, which is exhaled

The Bohr Effect

- The binding of O_2 to hemoglobin decreases with increasing H^+ concentration (lower pH) or when the hemoglobin is exposed to increased partial pressure of CO_2 (pCO_2).
- This phenomenon is known as Bohr effect.
- It is due to a change in the binding affinity of O_2 to hemoglobin
- Bohr effect causes a shift in the oxygen dissociation curve to the right

- **Bohr effect is primarily responsible for the release of O_2 from the oxyhemoglobin to the tissue.**
- **This is because of increased pCO_2 & decreased pH in the actively metabolizing cells**
- ***Binding of CO_2 forces the release of O_2.***

Carbonic anhydrase

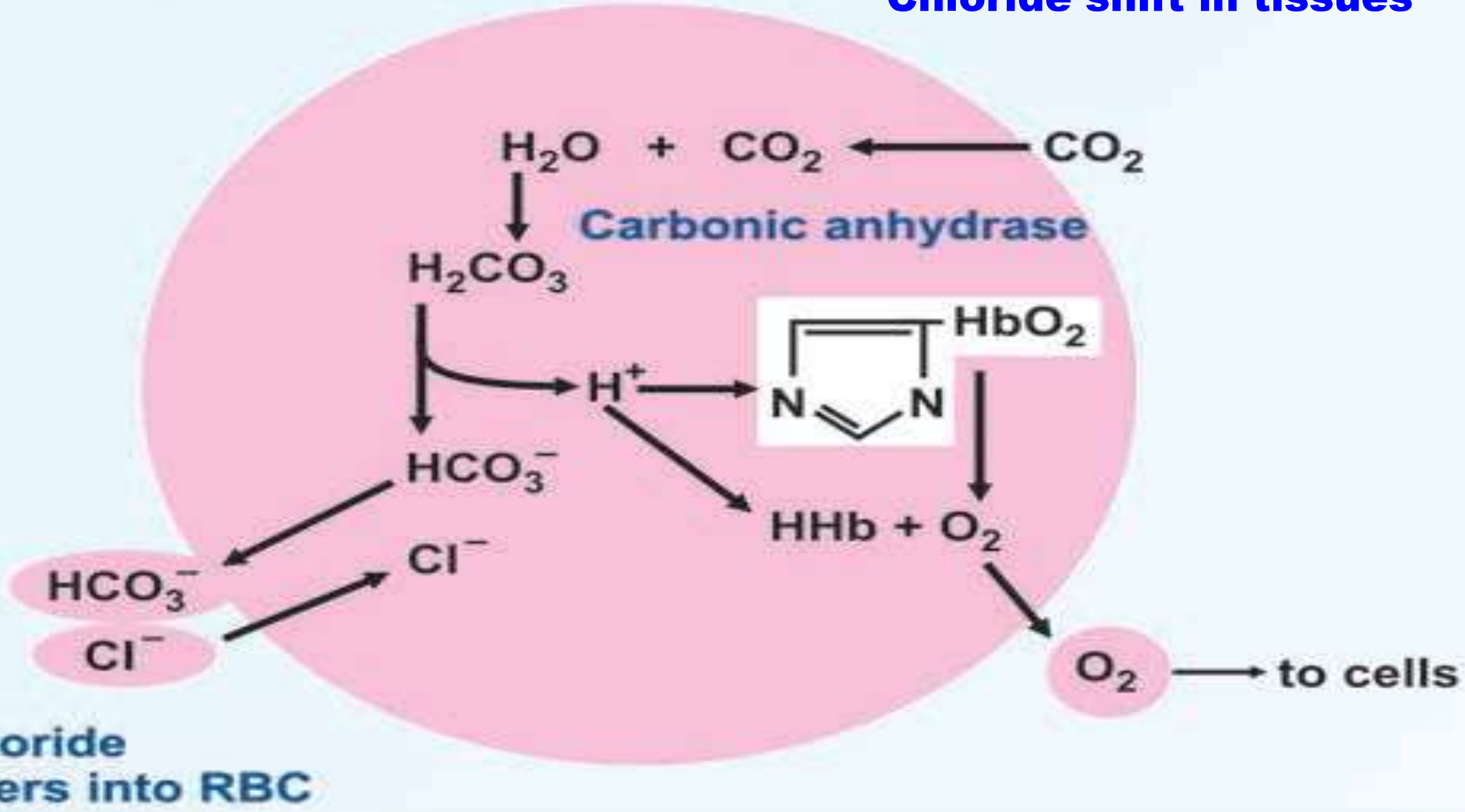
$$CO_2 + H_2O \longrightarrow H_2CO_3 \longrightarrow H^+ + HCO_3$$

- When carbonic acid ionizes, the intracellular pH falls.
- The affinity of Hb for O_2 is decreased & O_2 is unloaded to the tissues.

The Chloride Shift

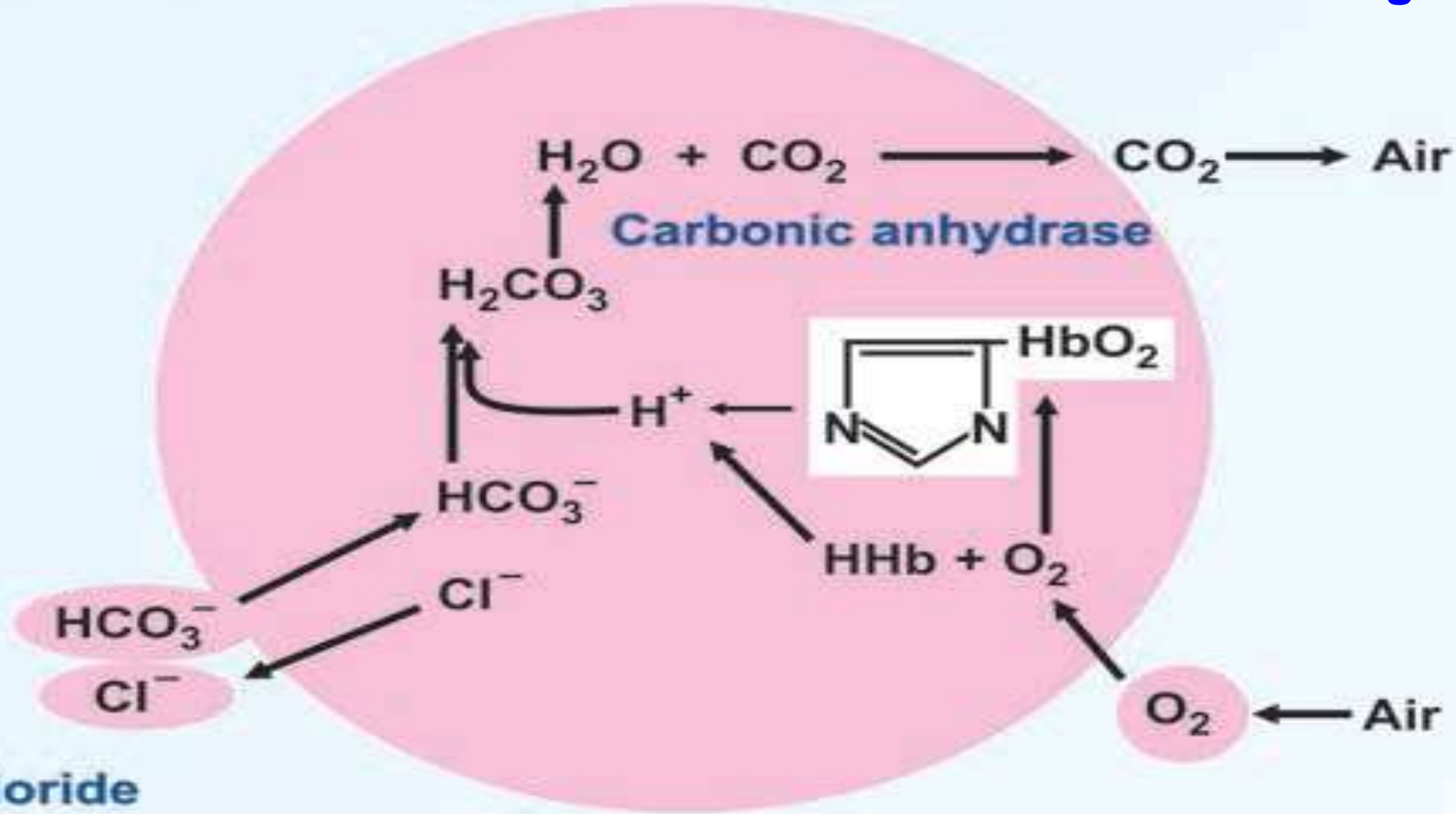
- When CO_2 is taken up, the HCO_3^- concentration within the cell increases.
- This would diffuse out into the plasma.
- Chloride ions from the plasma enter into cell to establish electrical neutrality.
- This is called chloride shift or Hamburger effect.
- RBCs are slightly bulged due to the increased chloride ions

Chloride shift in tissues



- When the blood reaches the lungs, the reverse reaction takes place.
- The deoxyhemoglobin liberates protons (H^+).
- These H^+ combine with HCO_3^- to form H_2CO_3.
- H_2CO_3 dissociated to CO_2 & H_2O by the carbonic anhydrase.
- The CO_2 is expelled.
- HCO_3^- binds H^+, more HCO_3^- from plasma enters the cell & Cl^- gets out (reversal of chloride shift)

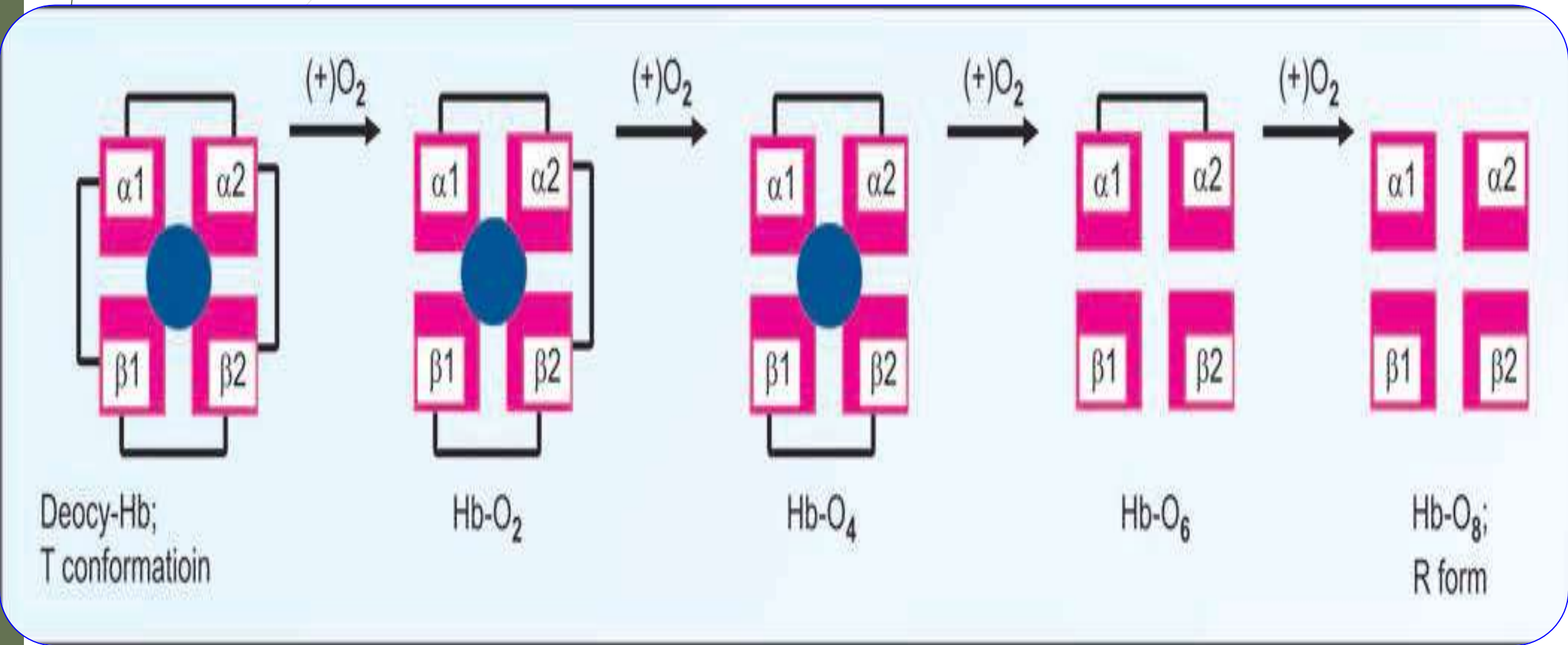
Chloride shift in lungs



Effect of 2,3-BPG

- 2,3-Bisphosphoglycerate is the most abundant organic phosphate in the erythrocyte.
- The 2,3-BPG is produced from 1,3-BPG, an intermediate of glycolytic pathway
- This short pathway, referred to as Rapaport-Leubering cycle
- The 2,3-BPG, binds to deoxy-Hb (and not to oxyhemoglobin) & decreases the O_2 affinity to Hb & stabilizes the T conformation.

As oxygen is added, salt bridges are successively broken and finally 2,3-BPG is expelled. Simultaneously the T (taught) confirmation of deoxy-Hb is changed into R(relaxed) confirmation of oxy-Hb.
Blue circle represents 2,3-bisphosphoglycerate (BPG)



- When the T form reverts to the R conformation, the 2,3-BPG is ejected.
- The reduced affinity of O_2 to Hb facilitates the release o f O_2 at the partial pressure found in the tissues.
- 2,3-BPC shifts the oxygen dissociation curve to the right
- The high oxygen affinity of fetal blood (HbF) is due to the inability of gamma chains to bind 2,3-BPG.

Mechanism of action of 2,3-BPG

- **One molecule of 2,3-BPG binds with one molecule (tetramer) of deoxyhemoglobin in the central cavity of the four subunits.**
- **This central pocket has positively charged (e.g. histidine, lysine) two β-globin chains.**
- ***Ionic bonds (salt bridges) are formed between the positively charged amino acids (of β-globins) with the negatively charged phosphate groups of 2,3-BPG***

- **Binding of 2,3-BPG stabilizes the deoxygenated hemoglobin (T-form) by crosslinking the β -chains**
- **On oxygenation of hemoglobin, 2,3-BPG is expelled from the pocket and the oxyhemoglobin attains the R-form of structure**

Clinical significance of 2,3-BPG

- **In hypoxia:**
- **The 2,3-BPG in erythrocytes is elevated in chronic hypoxic conditions associated with difficulty in O_2 supply.**
- **These include adaptation to high altitude, obstructive pulmonary emphysema**
- **In anemia:**
- **2,3-BPC levels are increased in severe anemia in order to cope up with the oxygen demands of the body.**

- This is an adaptation to supply as much O_2 as possible to the tissue, despite the low hemoglobin levels.
- In blood transfusion:
- Storage of blood in acid citrate-dextrose medium results in the decreased concentration of 2,3-BPG.
- Such blood when transfused fails to supply O_2 to the tissues immediately.

- **Addition of inosine (hypoxanthine-ribose) to the stored blood prevents the decrease of 2,3-BPG.**
- **The ribose moiety of inosine gets phosphorylated & enters the hexose monophosphate pathway and finally gets converted to 2.3-BPG**

Heme Synthesis

- Heme is the most important porphyrin containing compound.
- It is primarily synthesized in the liver & the erythrocyte-producing cells of bone marrow (erythroidc ells).
- Heme synthesis also occurs to some extent in other tissues.
- Mature erythrocytes lacking mitochondria are a notable exception.

Formation of δ-aminolevulinate

- **Glycine, a non-essential amino acid & succinyl CoA, an intermediate in the citric acid cycle, are the starting materials for porphyrin synthesis.**
- **Glycine combines with succinyl CoA to form δ –aminolevulinate (ALA).**
- **Catalysed by a PLP dependent δ –aminolevulinate synthase occurs in the mitochondria.**
- **It is a rate-controlling step in porphyrin synthesis**

Synthesis of porphobilinogen

- **Two molecules of δ -aminolevulinate condense to form porphobilinogen (PBG) in the cytosol.**
- **Catalysed by a Zn-containing enzyme ALA dehydratase.**
- **It is sensitive to inhibition by heavy metals such as lead.**

Formation of porphyrin ring

- **Porphyrin synthesis occurs by condensation of four molecules of porphobilinogen (PBG).**
- **The four pyrrole rings in porphyrin are interconnected by methylene (-CH_2) bridges derived from α- carbon of glycine.**
- **The interaction of two enzymes-namely uroporphyrinogen I synthase & uroporphyrinogen III cosynthase-results in condensation of porphobilinogen followed by ring closure & isomerization to produce uroporphyrinogen III.**

Conversion of uroporphyrinogen III to protoporphyrin IX

- Uroporphyrinogen decarboxylase decarboxylates all the four acetate (A) side chains to form methyl groups (M), to produce coproporphyrinogen.
- Coproporphyrinogen oxidase converts (oxidative decarboxylation) two of the propionate side chains (P) to vinyl groups (V) & results in the formation of protoporphyrinogen

- **Protoporphyrinogen oxidase oxidizes methylene groups (-CH_2-) interconnecting pyrrole rings to methenyl groups (=CH-).**
- **This leads to the synthesis of protoporphyrin IX.**

Synthesis of heme from protoporphyrin IX

- The incorporation of ferrous iron (Fe^{2+}) into protoporphyrin IX is catalysed by the enzyme ferrochelatase or heme synthetase.
- This enzyme can be inhibited by lead.

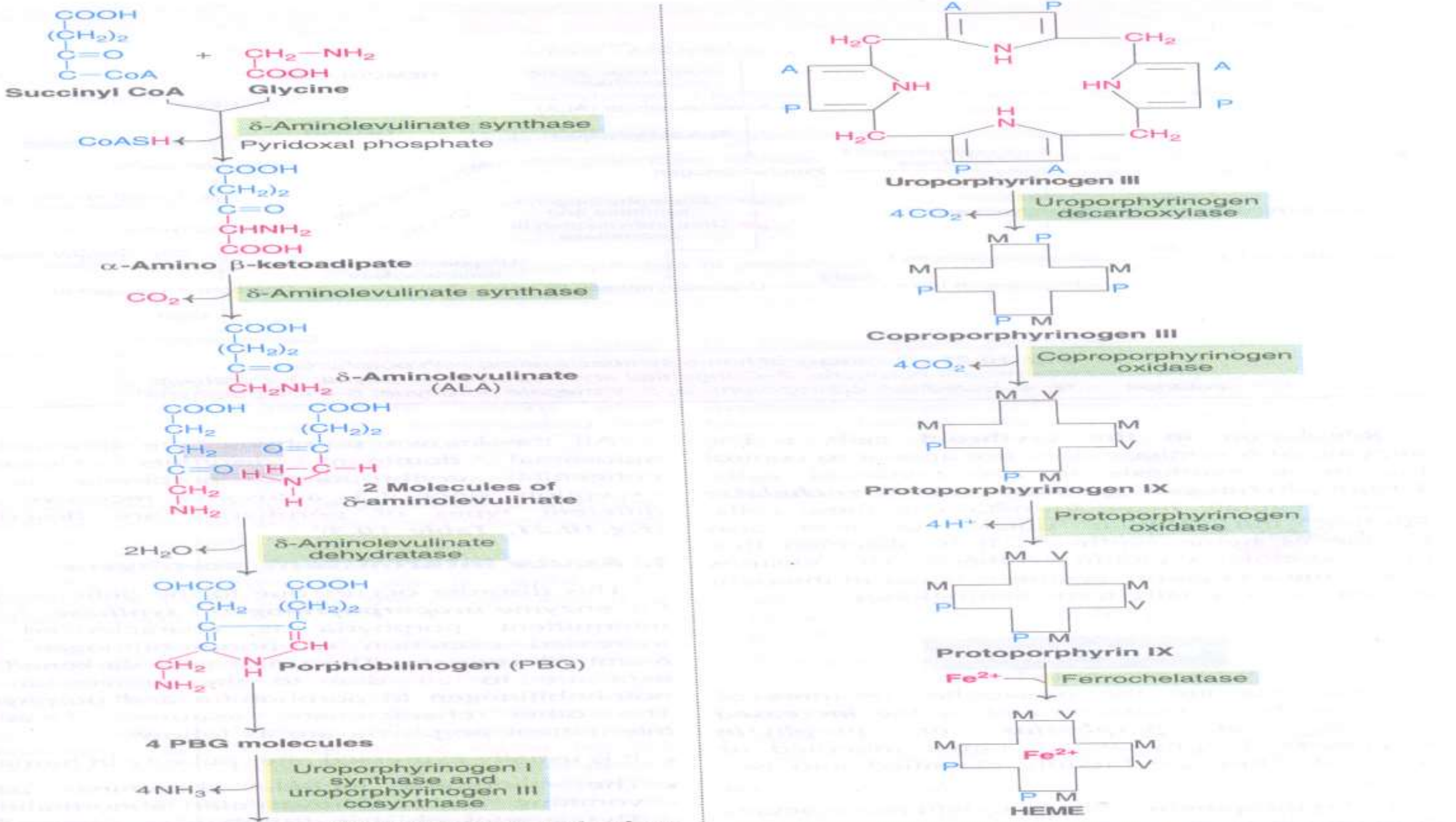


Fig. 10.20 contd. next column

Fig. 10.20 : *Biosynthesis of heme [A–Acetyl (— CH_2 — COO^-); P–Propionyl (— CH_2 — CH_2 — COO^-); M–Methyl (— CH_3); V–Vinyl (— CH = CH_2)].*

Regulation of heme synthesis

- Heme production in the liver is required for the formation of hemoproteins (e.g. Cytochrome P_{450} involved in detoxification) while in the erythroid cells, it is necessary for the synthesis of hemoglobin.
- Two different mechanisms exist for the regulation of heme biosynthesis in the liver & the erythroid cells

Regulation in the liver

- The first committed step in heme biosynthesis catalysed by δ-aminolevulinate (ALA) synthase is regulatory.
- Heme or its oxidized product hemin (Fe^{3+}) controls this enzyme activity by three mechanisms
- Feedback inhibition
- Repression of ALA synthatase
- Inhibition of transport of ALA synthase from cytosol to mitochondria (the site of action).

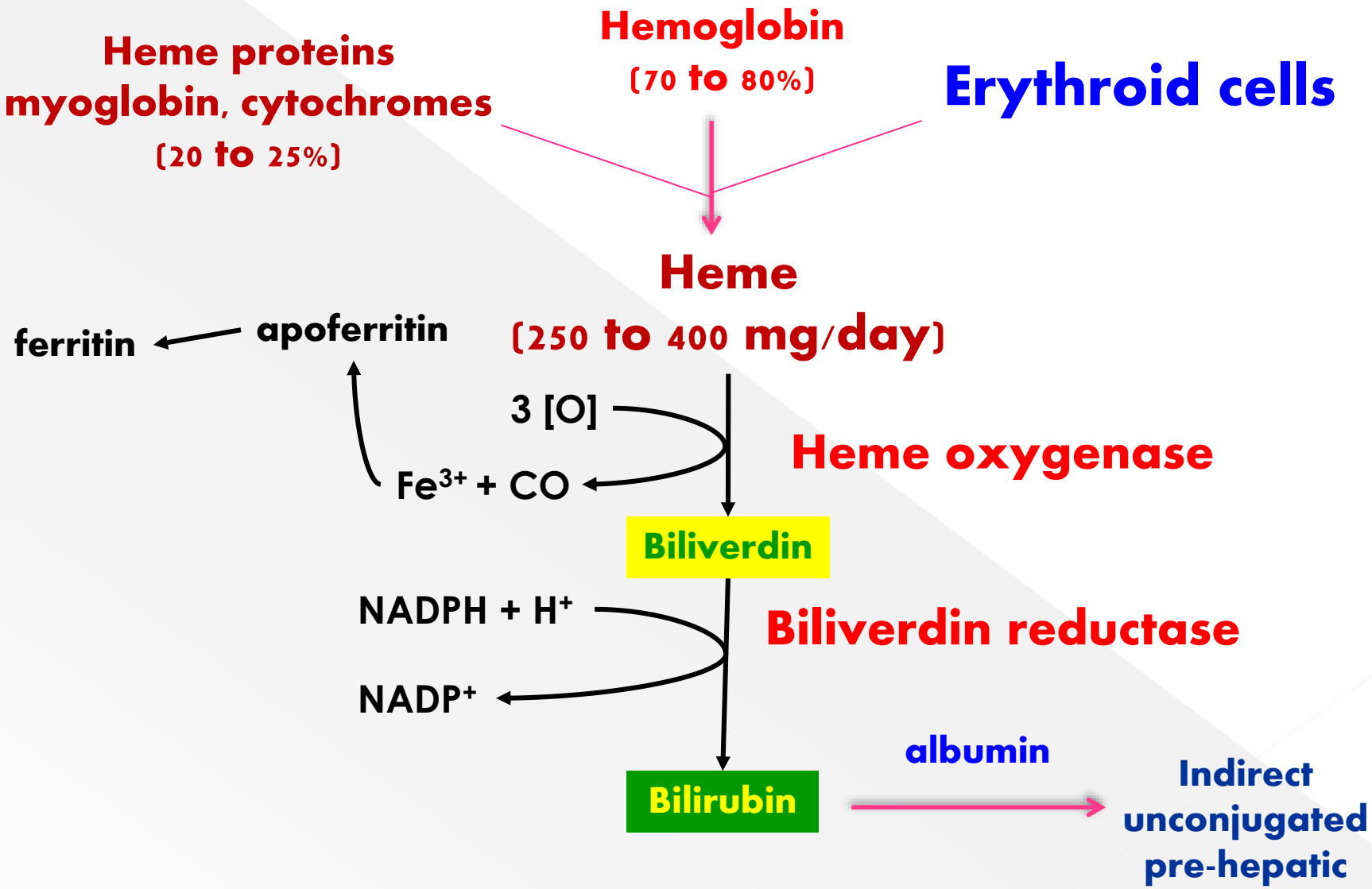
Effect of drugs on ALA synthase activity

- The activity of ALA synthase is markedly increased by the administration of a large number of drugs e.g. phenobarbital, insecticides, carcinogens etc.
- These compounds are mostly metabolized by a heme containing protein, cytochrome P_{450}
- On administration of drugs, cellular levels of heme are depleted due to its increased incorporation into cytochrome P_{450}.

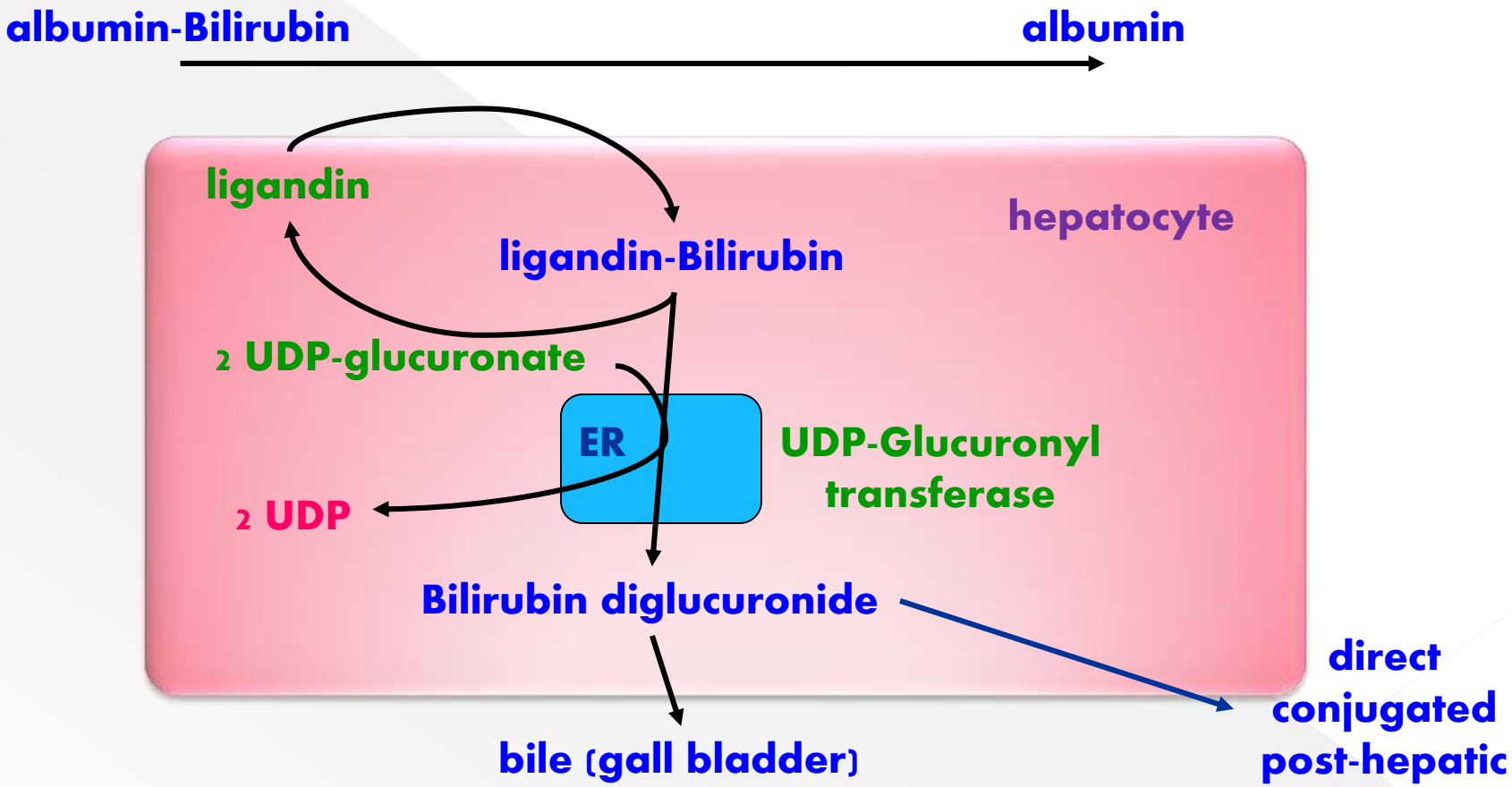
- **The reduced heme concentration increases the synthesis (derepression) of ALA synthase to meet the cellular demands.**
- **Regulation in the erythroid cells:**
- **The enzyme ALA synthase does not appear to control the heme synthesis in the erythroid cells.**
- **Uroporphyrinogen synthase & ferrochelatse mostly regulate heme formation in these cells**

Jaundice

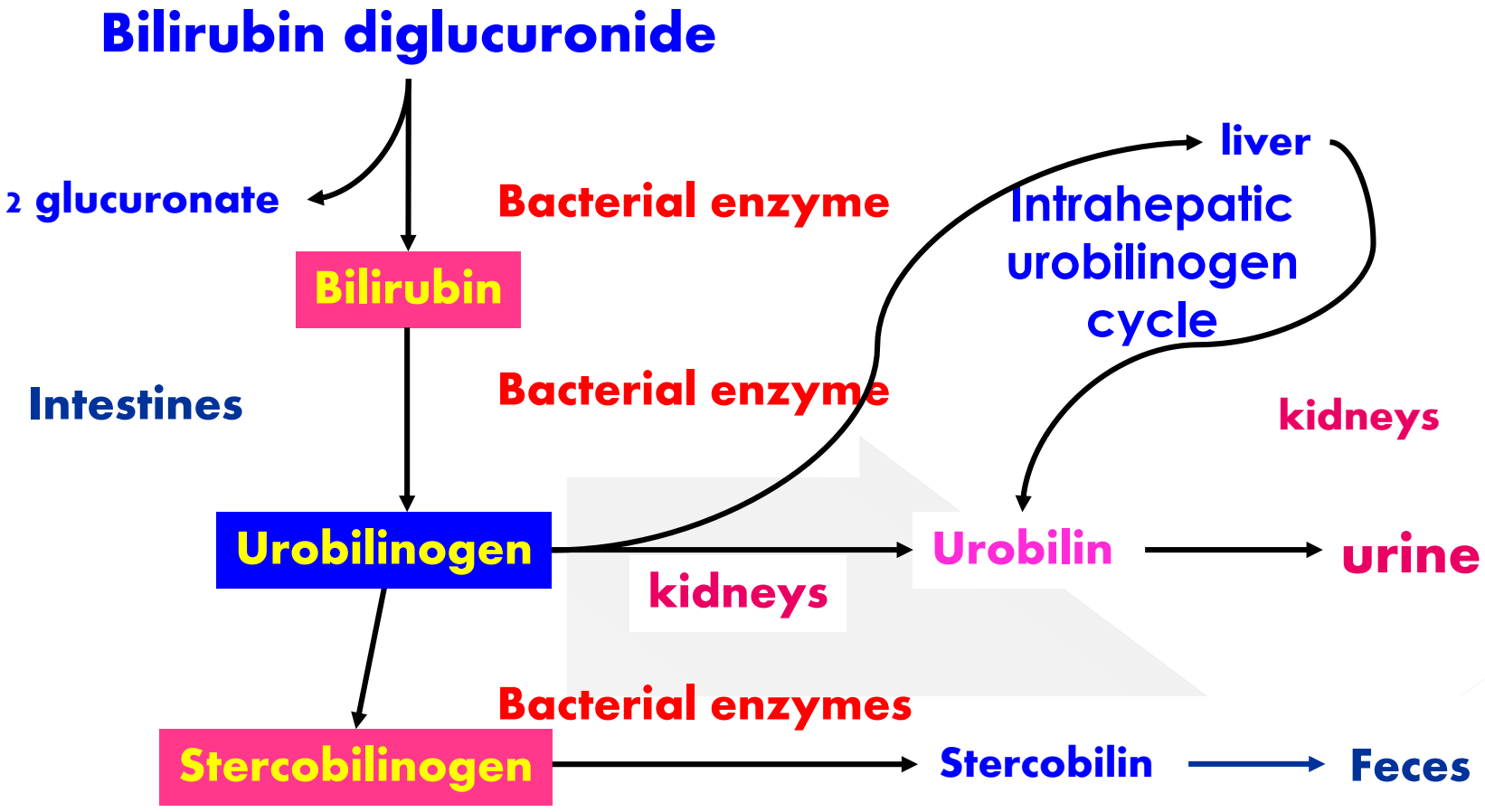
Bilirubin Production



Bilirubin Processing



Bilirubin Excretion



Plasma Bilirubin

- **Normal plasma bilirubin: 0.2–0.8 mg/dl.**
- **Unconjugated bilirubin: 0.2–0.6 mg/dl.**
- **Conjugated bilirubin: 0–0.2 mg/dl.**
- **If the plasma bilirubin level exceeds 1mg/dl, the condition is called hyperbilirubinemia.**
- **Levels between 1 & 2 mg/dl are indicative of latent jaundice.**

- When the bilirubin level exceeds 2 mg/dl, it diffuses into tissues producing yellowish discoloration of sclera, conjunctiva, skin & mucous membrane resulting in jaundice.
- Icterus is the Greek term for jaundice.

Van den Bergh Test for Bilirubin

- It is a specific test for for identificaion of increased serum bilirubin levels.
- Normal serum gives a negative van den Bergh reaction.
- Mechanism of the reaction:
- Van den Bergh reagent is a mixture of equal volumes of sulfanilic acid (in dilute HCl)& sodium nitrite.

- **Principle:**
- **Diazotised sulfanilic acid reacts with bilirubin to form a purple coloured azobilirubin.**
- **Direct and indirect reactions:**
- **Bilirubin as such is insoluble in water while the conjugated bilirubin is soluble.**

- Van den Bergh reagent reacts with conjugated bilirubin & gives a purple colour immediately (normally within 30 seconds.
- This is direct positive van den Bergh reaction.
- Addition of methanol (or alcohol) dissolves the unconjugated bilirubin & gives the van den Bergh reaction (normally within 30 minutes) positive.
- This is indirect positive.

- If the serum contains both unconjugated and conjugated bilirubin in high concentration, the purple colour is produced immediately (direct positive) which is further intensified by the addition of alcohol (indirect positive).
- This type of reaction is known as biphasic.

Van den Bergh reaction & jaundice

- Useful in understanding the nature of jaundice.
- This is due to jaundice is characterized by increased serum concentration of unconjugated bilirubin (hemolytic), conjugated bilirubin (obstructive) or both of them (hepatic).

- **Indirect positive** - **Hemolytic jaundice**
- **Direct positive** - **Obstructive jaundice**
- **Biphasic** - **Hepatic jaundice**
- **Bilirubin in urine:**
- **The conjugated bilirubin, being water soluble, is excreted in urine.**
- **Unconjugated bilirubin is not excreted.**
- **Bilirubin in urine can be detected by Fouchet's test or Gmelin's test.**

Hyperbilirubinemias

- Depending on the nature of the bilirubin elevated,
- Conjugated or Unconjugated hyperbilirubinemia.
- Based on the cause:
- Classified into congenital & acquired.

Congenital Hyperbilirubinemias

- They result from abnormal uptake, conjugation or excretion of bilirubin due to inherited defects.
- Crigler-Najjar Syndrome:
- Enzyme deficiency: UDP glucuronyl transferase.
- There is a defect in the conjugation.

- Type 1(Congenital non-hemolytic jaundice),
- There is severe deficiency of UDP glucuronyl transferase.
- The disease is often fatal & the children die before the age of 2.
- Jaundice usually appears within the first 24 hours of life.
- Unconjugated bilirubin level increases to more than 20 mg/dl, & results in kernicterus.

- **Type 2 disease: It is a milder form.**
- **Only the second stage of conjugation is deficient.**
- **When barbiturates are given, some response is seen & jaundice improves.**
- **Bilirubin level in blood exceeds 20 mg/dl in Crigler-Najjar syndrome Type 1**
- **Does not exceed 20 mg/ dl in Crigler-Najjar syndrome Type 2.**

Gilbert's Disease

- It is inherited as an autosomal dominant trait.
- The defect in uptake of bilirubin by the liver.
- Also due to reduced glucuronyl transferase activity
- Bilirubin level is usually around 3 mg/dl & patient is asymptomatic.
- Presence of mild jaundice.

Dubin-Johnson Syndrome

- It is an autosomal recessive trait.
- Defective excretion of conjugated bilirubin
- Conjugated bilirubin is increased in blood.
- The disease results from the defective ATP dependent organic anion transport in bile canaliculi.
- The bilirubin is deposited in the liver & the liver appears black.
- This is called as Black liver jaundice.

Rotor Syndrome

- It is an autosomal recessive condition.
- Bilirubin excretion is defective.
- There is no staining of the liver.

Acquired Hyperbilirubinemias

- Jaundice (also known as icterus) may be considered as a symptom rather than a disease.
- It is frequently caused due to multiple factors.
- Jaundice is 3 major types-
- Hemolytic Jaundice
- Hepatic Jaundice
- Obstructive Jaundice

Hemolytic Jaundice

- This condition is associated with increased hemolysis of erythrocytes (e.g. incompatible blood transfusion, malaria, sickle-cell anemia).
- This results in the overproduction of bilirubin beyond the ability of the liver to conjugate & excrete.

- **In hemolytic jaundice, more bilirubin is excreted into the bile leading to the increased formation of urobilinogen & stercobilinogen.**
- **Hemolytic jaundice is characterized by**
- **Elevation in the serum unconjugated bilirubin.**
- **Increased excretion of urobilinogen in urine.**
- **Dark brown colour of feces due to high content of stercobilinogen.**
- **Deposited in brain, leading to mental retardation, fits, toxic encephalitis & spasticity.**

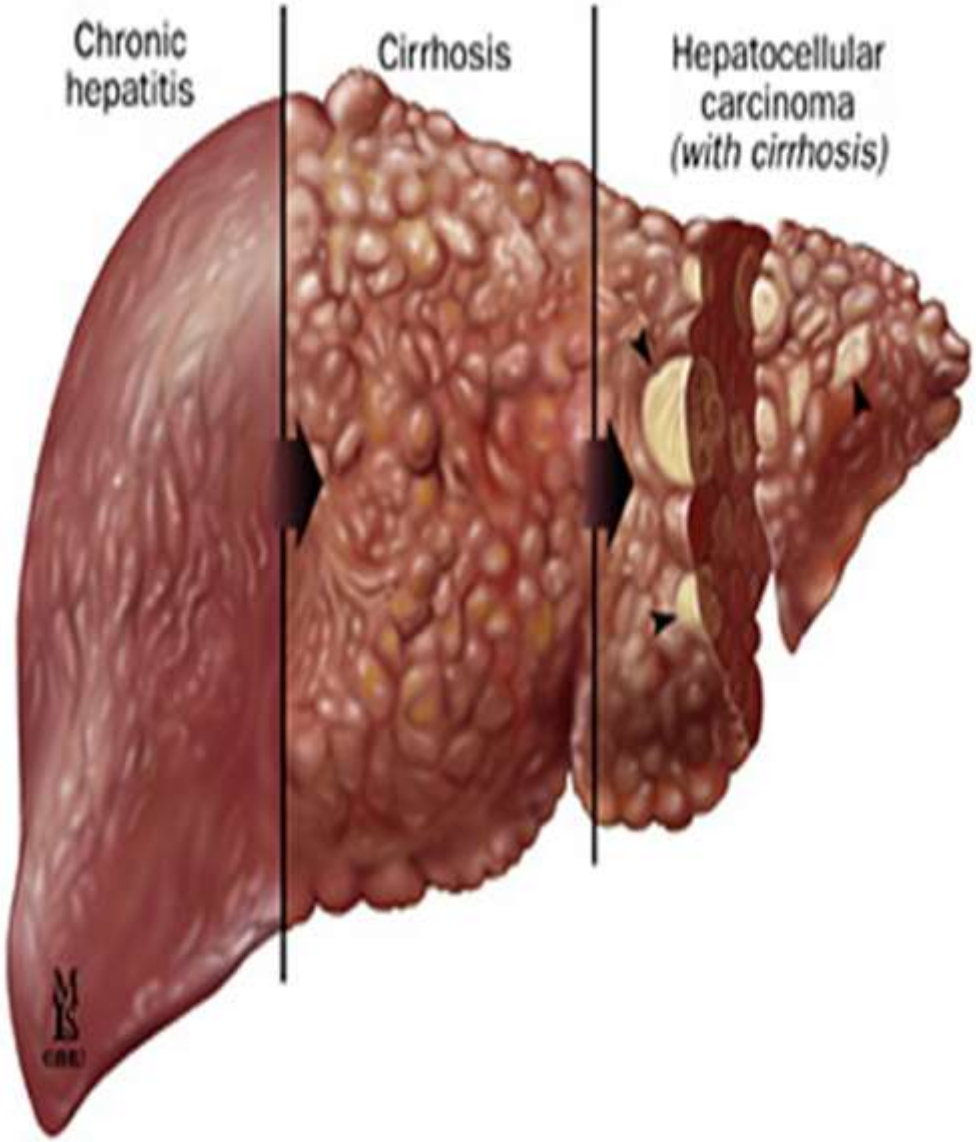
- If the child develops hemolytic disease, child may be given exchange transfusion along with phototherapy & barbiturates.
- *Phototherapy with blue light (440 nm wave length) isomerizes the insoluble bilirubin to more soluble isomers.*
- These can be excreted through urine without conjugation.

Hepatic (hepatocellular) jaundice

- It is caused by dysfunction of the liver due to damage to the parenchymal cells.
- This may be attributed to viral infection, poisons & toxins (chloroform, carbon tetrachloride, phosphorus etc.) cirrhosis of liver, cardiac failure etc.
- Among these, viral hepatitis is most common.

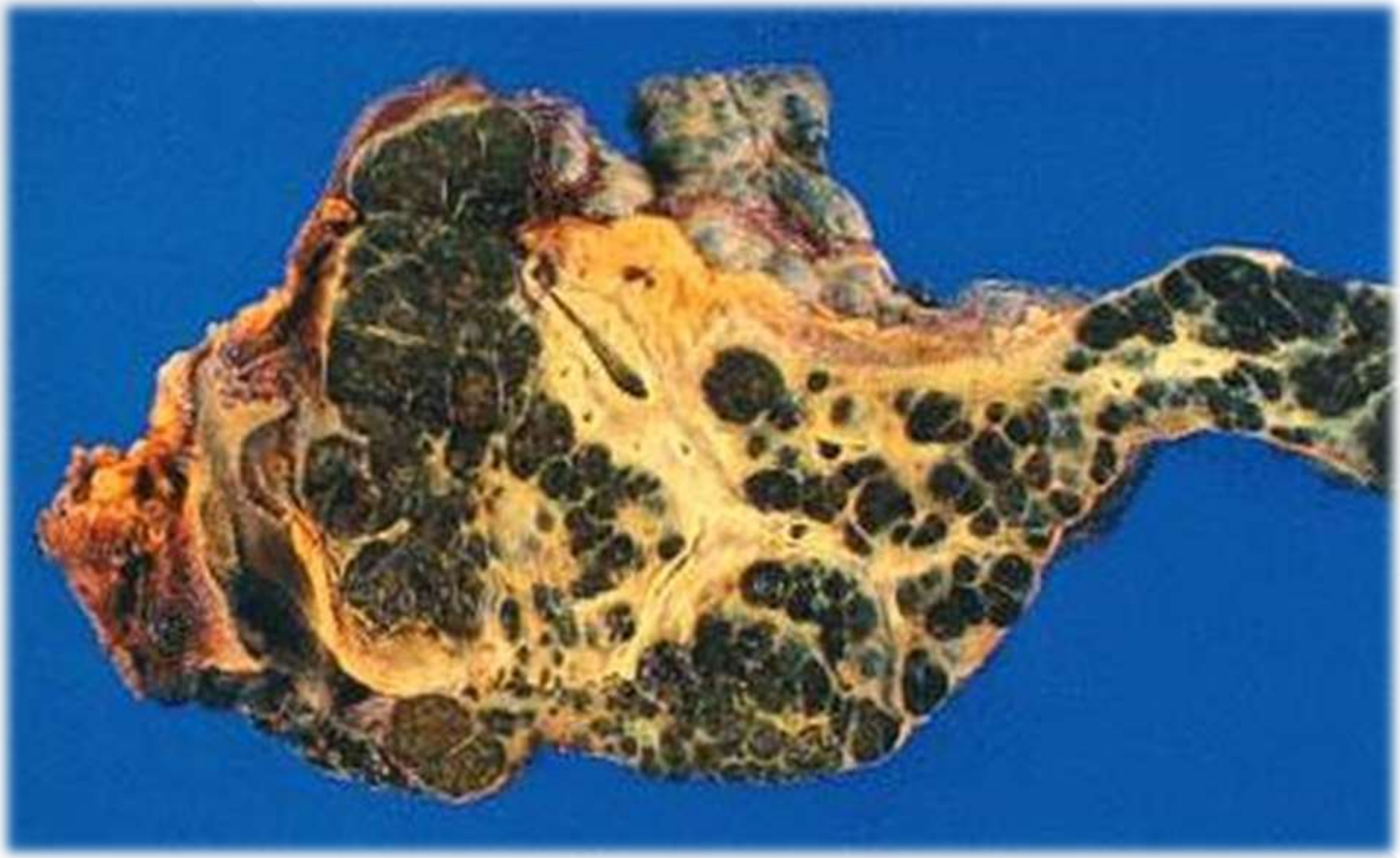
Hepatic jaundice





- Damage to the liver adversely affects the bilirubin uptake & its conjugation by liver cells.
- Hepatic jaundice is characterized by
- Increased levels of conjugated & unconjugated bilirubin in the serum.
- Dark coloured urine due to the excessive excretion of bilirubin & urobilinogen.

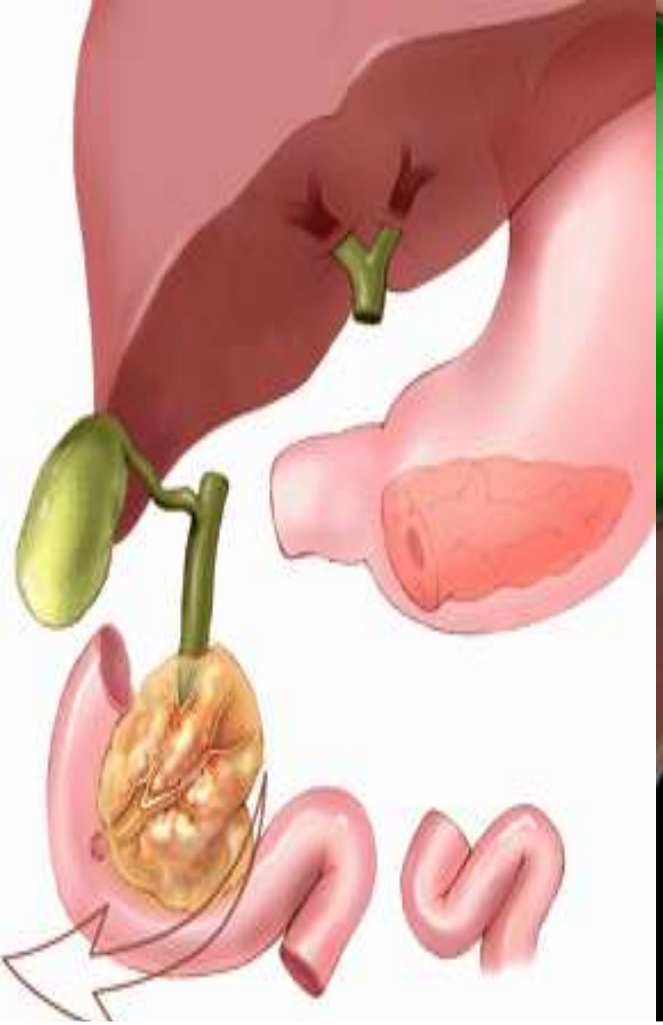
Gross specimen of cirrhosis of the liver

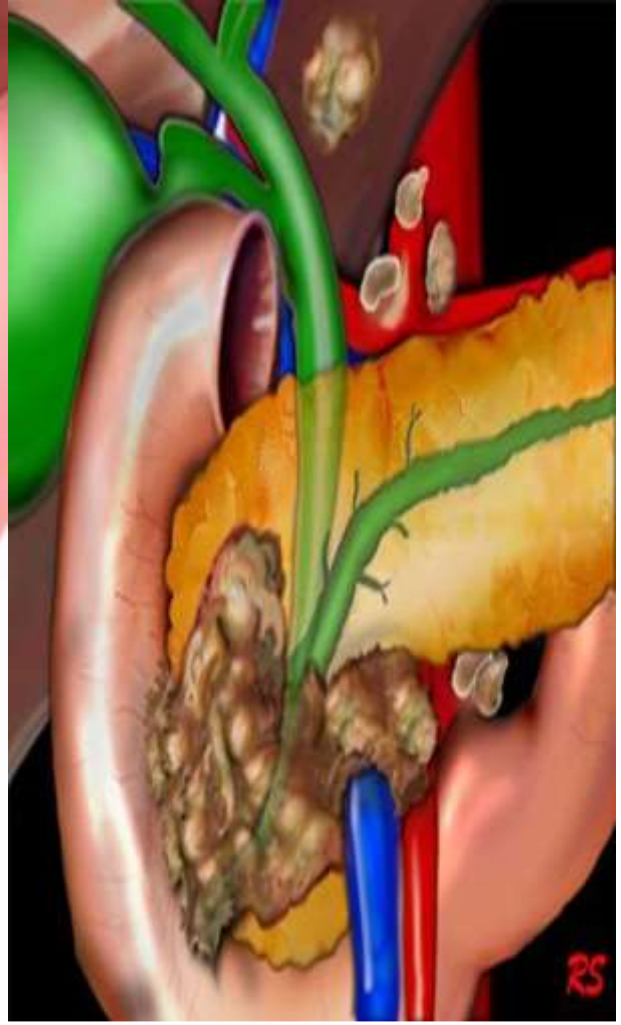


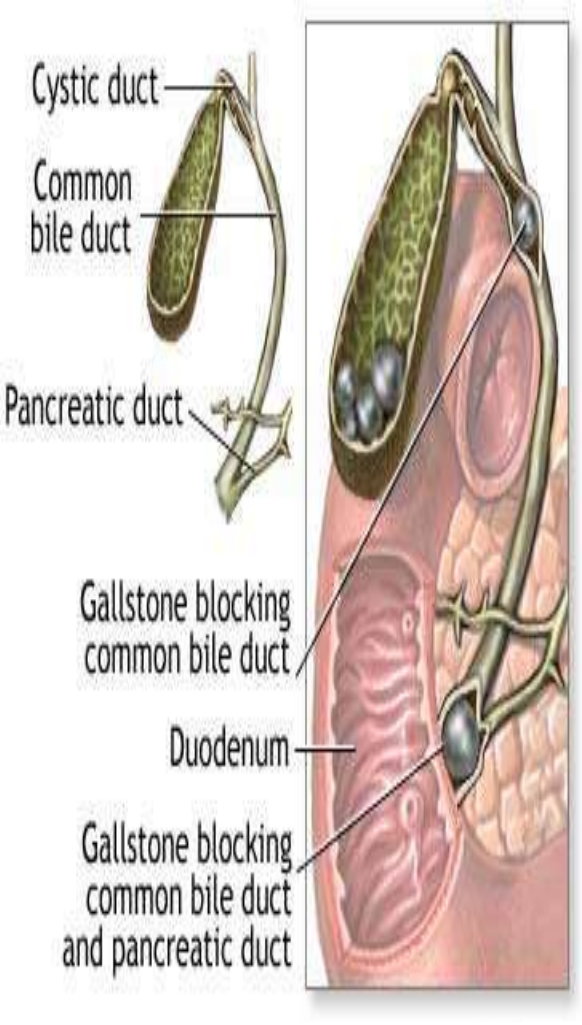
- Increased activities of alanine transaminase (SGPT) & aspartate transaminase (SGOT) released into circulation due to damage to hepatocytes.
- The patients pass pale, clay coloured stools due to the absence of stercobilinogen.
- Symptoms:
- Weakness, loss of appetite, hepatomegaly & nausea.

Obstructive Jaundice

- This is due to an obstruction in the bile duct that prevents the passage of bile into the intestine.
- The obstruction may be caused by gall stones, tumors etc.
- Due to the blockage in bile duct, the conjugated bilirubin from the liver enters the circulation.







Mechanism of Obstruction

- **Obstructive jaundice is characterized by**
- **Increased concentration of conjugated bilirubin in serum.**
- **Serum alkaline phosphatase is elevated as it is released from the cells of the damaged bile duct.**
- **Dark coloured urine due to increased excretion of bilirubin & clay coloured feces due to absence of stercobilinogen.**

- **Feces contain excess fat indicating impairment in fat digestion and absorption in the absence of bile (specifically bile salts).**
- **The patients experience nausea and gastrointestinal pain.**

Differential diagnosis of jaundice

	Hemolytic	Hepatic	Obstructive
Blood, free bilirubin	Increased	Increased	Normal
Blood,conj. bilirubin	Normal	Increased	Increased
Blood, ALP	Normal	Increased	Very high
Urine, bile salts	Nil	Nil	Present
Urine, conj.bilirubin	Nil	Nil	Present
Urine, bilinogens	Increased	Nil	Nil
Fecal urobilinogen	Increased	Decreased	Absent

Neonatal - physiological jaundice

- It is caused by increased hemolysis coupled with immature hepatic system for the uptake, conjugation & secretion of bilirubin.
- The activity of the enzyme UDP-glucuronyl transferase is low in the newborn.
- There is a limitation in the availability of the substrate UDP-glucuronic acid for conjugation.

- The serum uncojugated bilirubin is highly elevated (may go beyond 25 mg/dl).
- Which can cross the blood brain barrier.
- This results in hyperbilirubinemic toxic encephalopathy or kernicterus that causes mental retardation.

- The drug phenobarbital is used in the treatment of neonatal jaundice, as it can induce bilirubin metabolising enzymes in liver.
- In some neonates, blood transfusion may be necessary to prevent brain damage.

Phototherapy

- Bilirubin can absorb blue light (420-470 nm)
- Phototherapy deals with the exposure of the jaundiced neonates to blue light.
- By a process called photoisomerization, the toxic native unconjugated bilirubin gets converted into a non-toxic isomer namely lumirubin.

- **Lumirubin can be easily excreted by the kidneys in the unconjugated form (in contrast to bilirubin which cannot be excreted).**
- **Serum bilirubin is monitored every 12-24 hours, and phototherapy is continuously carried out till the serum bilirubin becomes normal (< 1 mg/dl)**

Phototherapy



Breast milk jaundice

- In some breast-fed infants, prolongation of the jaundice has been attributed to high level of an estrogen derivative in maternal blood, which is excreted through the milk.
- This would inhibit the glucuronyl transferase
- Sulpha & other drugs may release bilirubin from albumin, & may cause jaundice in newborn.

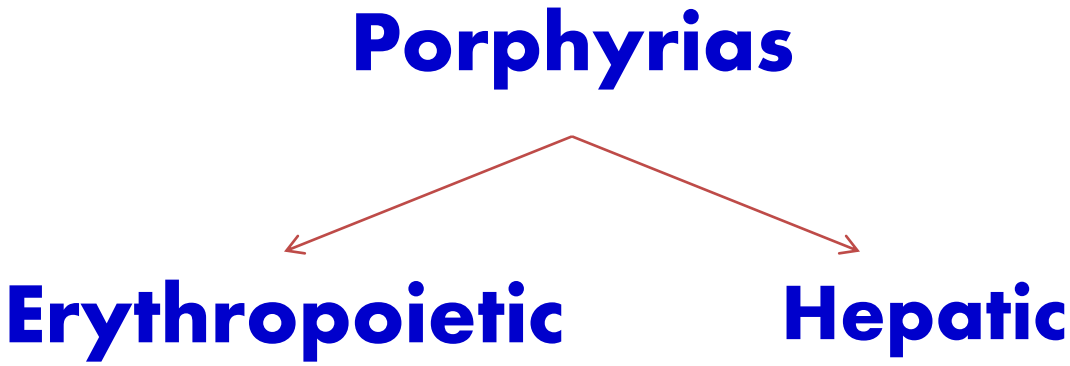
Thank you

Porphyrias

Porphyrias

- Porphyrias are the metabolic disorders of heme synthesis, characterized by the increased excretion of porphyrins or porphyrin precurcors.
- Porphyrias are either inherited or acquired.
- The most common acquired form of porphyria is due to lead poisoning.

Classification of porphyrias



- **Erythropoietic (bone marrow):**
- **Enzyme deficiency occurs in the erythrocytes.**
- **Hepatic:**
- **Enzyme defect lies in the liver.**

Different types of porphyrias

Type of porphyria	Enzyme defect	Characteristics
HEPATIC		
Acute intermittent porphyria	Uroporphyrinogen I synthase	Abdominal pain, neuropsychiatric symptoms
Porphyria cutanea tarda	Uroporphyrinogen decarboxylase	Photosensitivity
Hereditary coproporphyria	Coproporphyrinogen oxidase	Abdominal pain, Photosensitivity , neuropsychiatric symptoms
Variegate porphyria	Protoporphyrinogen oxidase	Abdominal pain, Photosensitivity , neuropsychiatric symptoms
ERYTHROPOIETIC PORPHYRIA		
Congenital erythropoietic porphyria	Uroporphyrinogen III cosynthase	Photosensitivity , increased hemolysis
Protoporphyria	Ferrochelatase	Photosensitivity

Hepatic porphyria

- **Acute intermittent porphyria:**
- **Enzyme defect: Uroporphyrinogen I synthase**
- **Characteristic features:**
- **Increased excretion of porphobilinogen & γ-ALA.**
- **Urine gets darkened on exposure to air due to conversion of porphobilinogen to porphobilin & porphyrin.**
- **It usually expressed after puberty.**

Symptoms

- **Abdominal pain, vomiting & cardiovascular abnormalities.**
- **Neuropsychiatric distrubances- due reduced activity of tryptophan pyrrolase (caused by depleted heme levels) resulting in the accumulation of tryptophan & 5-hydroxytryptamine.**

- Symptoms are more severe after administration of drugs (e.g. barbiturates)
- It induce the synthesis of cytochrome P_{450}.
- This is due to the increased activity of ALA synthase causing accumulation of PBG & ALA.
- These patients are not photosensitive.
- It is treated by administration of hematin, it inhibits ALA sytnthase & accumulation of porphobilinogen.

Porphyria cutanea tarda

- This is also known as cutaneous hepatic porphyria & is the most common porphyria.
- It associated with liver damage caused by alcohol overconsumption or iron overload.
- Enzyme deficiency:
- Uroporphyrinogen decarboxylase.

Characteristic features

- Increased excretion of uroporphyrins (I & III) & rarely porphobilinogen.
- Cutaneous photosensitivity is the most important clinical manifestation of these patients.
- Skin fragility, scarring, sclerodermoid skin changes.
- Liver exhibits fluorescence due to high concentration of accumulated porphyrins.





Porphyria cutanea tarda

Hereditary coproporphyria

- **Enzyme defect:**
- **Coproporphyrinogen oxidase.**
- **Coproporphyrinogen III & other intermediates (ALA and PBC) of heme synthesis prior to the blockade are excreted in urine & feces.**
- **The patients are photosensitive.**

Symptoms

- **Symptoms are similar to acute intermittent porphyria**
- **Abdominal pain, vomiting & cardiovascular abnormalities.**

- **Neuropsychiatric distrubances- due reduced activity of tryptophan pyrrolase (caused by depleted heme levels) resulting in the accumulation of tryptophan & 5-hydroxytryptamine.**
- **It is treated by administration of hematin, it inhibits ALA stnthase & accumulation of porphobilinogen.**

Variegate porphyria

- **Enzyme defect: Protoporphyrinogen oxidase**
- **Due to this blockade, protoporphyrin IX required for the ultimate synthesis of heme is not produced.**
- **Almost all the intermediates (porphobilinogen, coproporphyrin, uroporphyrin, protoporphyrin etc.) of heme synthesis accumulate in the body & are excreted in urine and feces.**
- **The urine of these patients is coloured & they exhibit photosensitivity**

Erythropoietic porphyria

- **Congenital erythropoietic porphyria:**
- **Enzyme defect:**
- **Uroporphyrinogen III cosynthase.**
- **Also caused by an imbalance between the activities of uroporphyrinogen I synthase and uroporphyrinogen III cosynthase**

Characteristic features

- It is a rare congenital disorder.
- Mostly contained in erythropoietic tissues (bone)
- Individuals excrete uroporphyrinogen I & coproporphyrinogen I which oxidize respectively to uroporphyrin I & coproporphyrin I.

- The patients are photosensitive (itching & burning of skin when exposed to light).
- Skin pain or burning in sunlight
- Erythema, swelling.
- Erosions in light exposed areas-mainly in face & hands.
- Scarring - shallow circular or linear.
- Waxy thickening of the skin.
- Increased hemolysis.



Erythropoietic porphyria

Protoporphyria

- Also known as erythropoietic protoporphyria.
- Enzyme defect:
- Ferrochelatase.
- Protoporphyrin IX accumulates in tissues & is excreted into urine & feces.
- Reticulocytes (young RBC) & skin biopsy exhibit red flourorescence.

Acquired or toxic porphyria

- It occur due to toxicity of several compounds.
- Exposure of the body to heavy metals (e.g. lead), toxic compounds (hexachlorobenzene) and drugs (e.g. griseofulvin) inhibits many enzymes in heme synthesis.
- These includes ALA dehydratase, uroporphyrin I synthase & ferrochelatase.

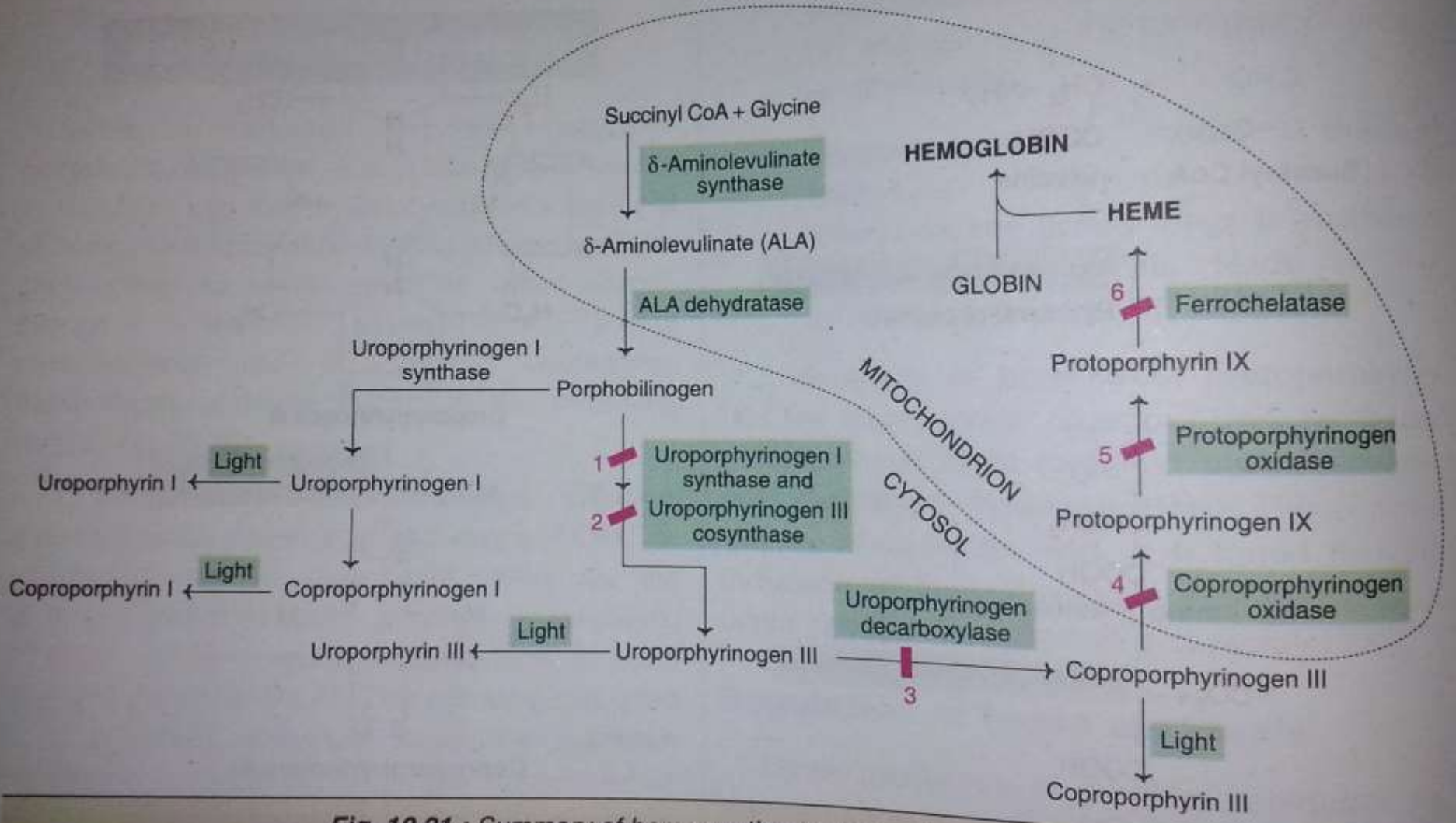


Fig. 10.21 : *Summary of heme synthesis along with porphyrias (1–Acute intermittent porphyria; 2–Congenital erythropoietic porphyria; 3–Porphyria cutanea tarda; 4–Hereditary coproporphyria; 5–Variegate porphyria; 6–*

Thank You